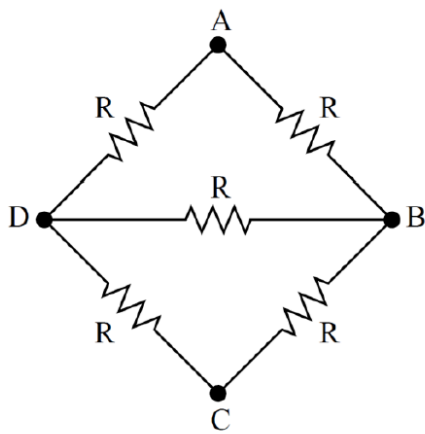
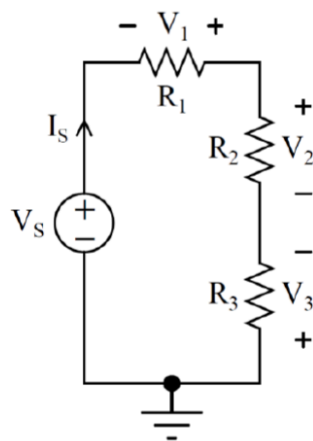


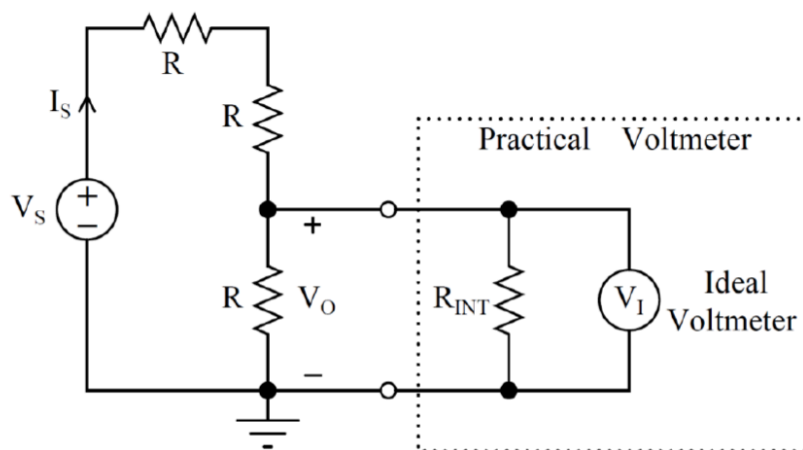
1 Lab 1 Calculations



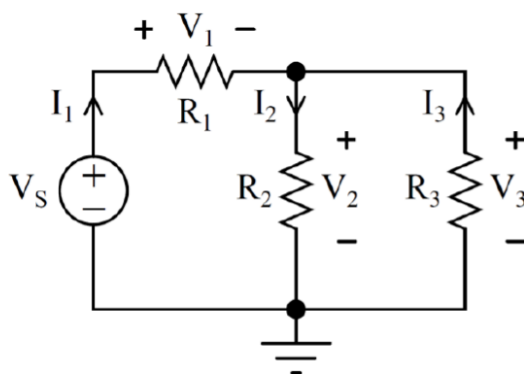
(a)



(b)



(c)



(d)

1.A Equivalent Resistance Calculation

1.A.1 Figure 1-6(a)

1.A.1.a

The equivalent resistance between nodes A & B, R_{AB}

$$\begin{aligned}
 R_{\text{EQ}_{\text{AB}}} &= R_{\text{AB}} \parallel [R_{\text{AD}} + (R_{\text{DC}} + R_{\text{CB}}) \parallel R_{\text{DB}}] \\
 &= 10k\Omega \parallel [10k\Omega + (10k\Omega + 10k\Omega) \parallel 10k\Omega] \\
 &= 10k\Omega \parallel [10k\Omega + (20k\Omega) \parallel 10k\Omega] \\
 &= 10k\Omega \parallel \left[10k\Omega + \frac{20}{3}k\Omega \right] \\
 &= 10k\Omega \parallel \left[\frac{50}{3}k\Omega \right] \\
 R_{\text{EQ}_{\text{AB}}} &= \frac{25}{4}k\Omega = 6.25k\Omega
 \end{aligned}$$

1.A.1.b

The equivalent resistance between nodes B & D, R_{BD}

$$\begin{aligned}
 R_{\text{EQ}_{\text{BD}}} &= R_{\text{BD}} \parallel (R_{\text{BA}} + R_{\text{AD}}) \parallel (R_{\text{BC}} + R_{\text{CD}}) \\
 &= 10k\Omega \parallel (10k\Omega + 10k\Omega) \parallel (10k\Omega + 10k\Omega) \\
 &= 10k\Omega \parallel (20k\Omega) \parallel (20k\Omega) \\
 &= 10k\Omega \parallel 10k\Omega \\
 R_{\text{EQ}_{\text{BD}}} &= 5k\Omega
 \end{aligned}$$

1.A.1.c

The equivalent resistance between nodes A & C, R_{AC}

$$R_{\text{EQ}_{\text{AC}}} = 10k\Omega???$$

1.B Voltage and Current Calculation

1.B.1 Figure 1-6(b)

1.B.1.a

Calculate V_1 , V_2 , V_3 and I_S WITHOUT using the voltage division rule.

Known/Chosen Values

$$\begin{aligned}V_s &= 9v \\ R_1 &= 10k\Omega \\ R_2 &= 20k\Omega \\ R_3 &= 30k\Omega\end{aligned}$$

KVL@ I_s

$$V_1 + V_3 + V_s = V_2$$

Ohm's Law@ R_1

$$V_1 = I_s \times R_1$$

Ohm's Law@ R_2

$$-V_2 = I_s \times R_2$$

Ohm's Law@ R_3

$$V_3 = I_s \times R_3$$

```
% Define symbolic variables
syms is v1 v2 v3

% Define equations
eq1 = v1 + v3 + 9 == v2;
eq2 = v1 == is * 10000;
eq3 = v2 == -is * 20000;
eq4 = v3 == is * 30000;

% Solve equations
sol = solve([eq1, eq2, eq3, eq4], [is, v1, v2, v3])
```

Solution

```
sol =
  is: -3/20000
  v1: -3/2
  v2: 3
  v3: -9/2
```

$$\begin{aligned}i_s &= -150\mu A \\ v_1 &= -1.5v \\ v_2 &= 3v \\ v_3 &= -4.5v\end{aligned}$$

1.B.1.b

Calculate V_1 , V_2 and V_3 again using the voltage division rule and compare the values with those determined in Part a.

1.B.1.c

Is the addition of V_S , V_1 and V_3 equal to the value of V_2 ? Why? Explain your reasoning in detail.

1.B.2 Figure 1-6(c)

1.B.2.a

Calculate V_O for $R = 10k\Omega$

```
% Define symbolic variables
syms vo

% Define equations
eq1 = (9 - vo)/20000 == (vo/10000) + (vo/10000000);

% Solve equations
sol = solve([eq1], [vo])
```

```
sol =
    4500/1501
```

1.B.2.b

Calculate V_O for $R = 10M\Omega$

```
% Define symbolic variables
syms vo

% Define equations
eq1 = (9 - vo)/20000000 == (vo/10000000) + (vo/100000000);

% Solve equations
sol = solve([eq1], [vo])
```

```
sol =
    9/5
```

1.B.3 Figure 1-6(d)

1.B.3.a

Calculate I_1 , I_2 , and I_3 WITHOUT using the current division rule.

KCL@ V_2

$$\frac{9 - V_2}{R_1} + \left(-\frac{V_2}{R_3}\right) = \frac{V_2}{R_2}$$
$$\frac{9 - V_2}{10k\Omega} + \left(-\frac{V_2}{30k\Omega}\right) = \frac{V_2}{20k\Omega}$$

```
% Define symbolic variables
syms i1 i2 i3 v2

% Define equations
eq3 = i1 + i3 == i2;
eq4 = i1 == 9-v2 / 10000;
eq5 = i2 == v2 / 20000;
eq6 = i3 == -v2 / 30000;

% Solve equations
sol = solve([eq3, eq4, eq5, eq6], [v2, i1, i2, i3])
```

```
sol =
    v2: 540000/11
    i1: 45/11
    i2: 27/11
    i3: -18/11
```

$$v_2 = 49.\overline{09}v$$

$$i_1 = 4.\overline{09}A$$

$$i_2 = 2.\overline{45}A$$

$$i_3 = -1.\overline{63}A$$

1.B.3.b

Calculate I2 and I3 again using the current division rule and compare the values with those determined in Part a.

1.B.3.c

Is the addition of I1 and I3 equal to the value of I2? Why? Explain your reasoning in detail.